

Classification Lecture 1

Thursday, 26 September 2024

6:06 AM

Classification is a type of supervised learning where the goal is to assign a label or class to an input based on its features.

e.g. spam detection, disease diagnosis, image recognition, etc.

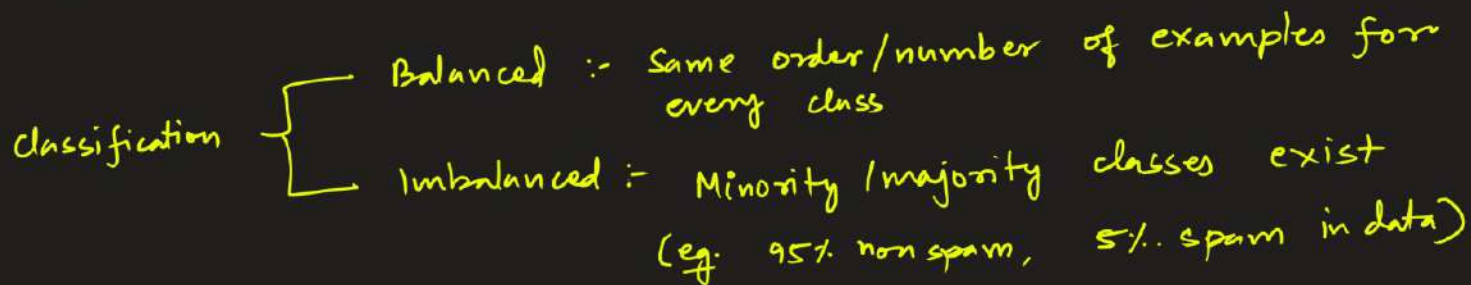
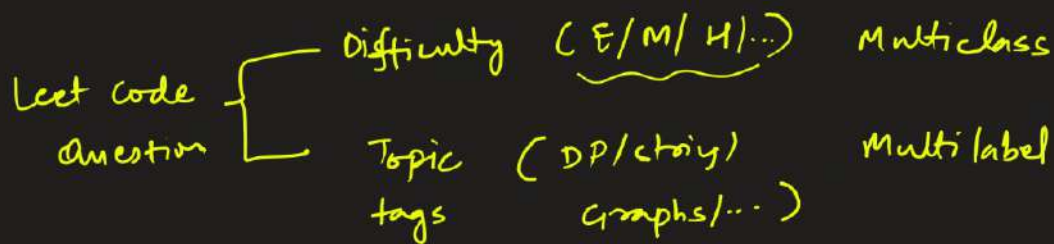
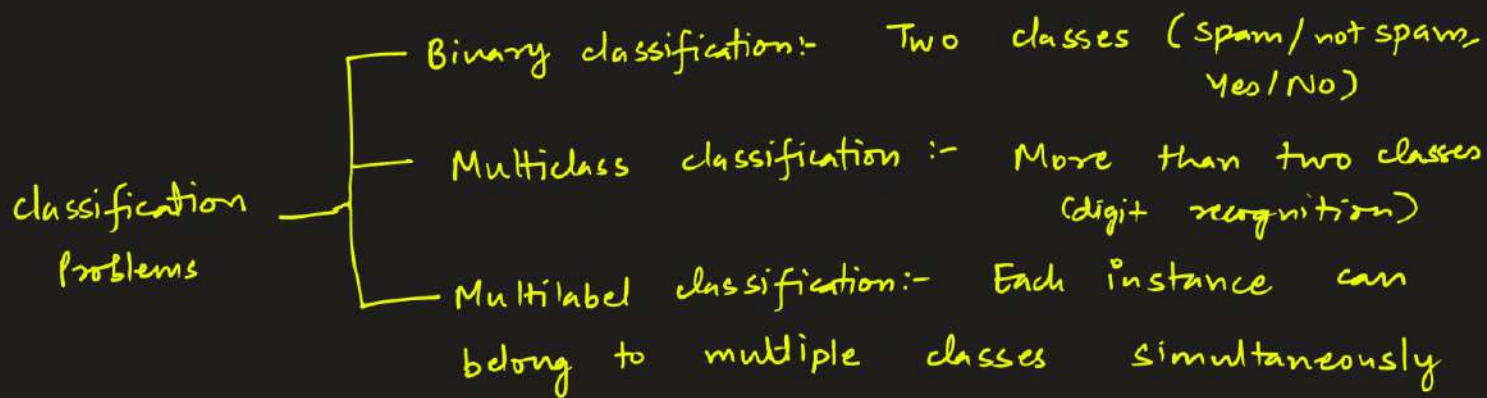
In classification, input is a set of features, and the output is a distinct category or class. The model learns a mapping between the features & the class labels based on the labeled dataset (supervised). It is a task of predicting categorical output/label for a given input.

Problem:- $D = \{(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)\}$,

x_i :- features (input)

y_i :- class label (output) associated with x_i

Objective:- learn a model $f(x)$ that can predict y for an unseen instance x .



Input interpretation in classification

Input
Features

Numerical features :- Represent the quantity or measurements and can take wide range of values
(eg. Age, Salary, height, temperature, etc.)

Categorical Features :- represent data that can be divided into categories or groups
(eg. Gender, color, Marital status)

Textual & Image Features :- Text or pixels of an image can be an input as well. These require special handling, like converting text into number vectors or processing pixel values in an image.

Image ↓ Photo x	Numerical ↓ Age	Textual ↓ Name x	Numerical ↓ Height x	Categorical ↓ Gender	Numerical ↓ Salary	Categorical ↓ Marital Status	y (Categorical) ↓ Likely to take a loan
	25	Lokesh	6'	Male	150000	Unmarried	Yes/No
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Handling Categorical Features

Categorical features cannot be directly used by most machine learning algorithms since they expect numerical inputs. Therefore categorical inputs must be encoded into numerical features.

Categorical data: —
 — Ordinal :- data can be ranked / ordered in a particular sequence
 — Nominal :- No order in the categories.

① Label Encoding: Each category is assigned a numerical (integer) value

eg: Red = 0
Green = 1
Blue = 2

less Confident = 1 ✓
 Slightly Confident = 2 ✓
 Very Confident = 3 ✓
 3 ✗
 7 ✗
 4 ✗

Pros:- Simple and works on ordinal data

Cons:- For nominal data, label encoding can create misleading ordinal relationships (eg Green > Red, Blue > Green)

② **One-Hot Encoding** :- creates binary features for each category, assigning 1 in the presence of category and 0 in the absence of the category.

eg:

	R	G	B
Red	[1, 0, 0]		
Green	[0, 1, 0]		
Blue	[0, 0, 1]		

Pros:- Eliminates the issues of ordinal relationships

Cons:- Increases the dimensionality and complexity of the input & the model.

eg. #	Age (Numerical)	Gender (M/F)	Subscription (No/Plus/Pro)	y Purchase
		M → <u>[1,0]</u>	Plus → <u>1</u>	Yes
<u>25</u> ← 25		F → <u>[0,1]</u>	Plus → <u>1</u>	No
60 ← 60		F → <u>[0,1]</u>	No → <u>0</u>	Yes
<u>13</u> ← 13		⋮	Pro → <u>2</u>	⋮

↓
Gender nominal
data (one hot
encoding)

↘
ordinal data
(label
encoding)

Handle Numerical data

Numerical features can be used directly by the models, but some preprocessing may still be required.

① Feature Scaling:-

└ Min-Max Scaling

$$x_{\text{scaled}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

└ Standardization

(Z-score normalization)

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma}$$

② Handle outliers

└ Clipping:- Restrict values above or below a threshold.

└ Log Transformation:- Reduce the impact of extreme values by compressing the scale of large numbers.

Dealing with Missing Data

- **Removal** :- Remove rows with an feature missing
Remove columns / features with a lot of data missing
→ can lead to loss of valuable information
- **Imputation** :- Involves filling the missing values with estimates, such as mean / mode / Median imputation.

Output interpretation in classification

In classification, the model's output can take various forms depending on the algorithm used.

→ **class probabilities** :- A probability distribution over possible class labels (Logistic Regression, Neural Networks ...)

eg. Spam :- 0.73, Not Spam :- 0.27

↙
Binary classification

single probability value

$p = 0.8$, other class :- $1 - 0.8 = 0.2$

→ **Multi-class classification**

probability distⁿ over categories

Cat :- 0.2 Dog :- 0.7 Bird :- 0.1

→ **class labels** :- A direct assignment to one of the possible classes (Decision Trees, K-NN, ...)

Thresholding

In classification, the output is often a probability value (e.g. 0.7) and a decision needs to be made about whether to classify the instance as 1 or 0.

$$\hat{y} = \begin{cases} 1, & \text{if } p \geq \text{threshold} \\ 0, & \text{if } p < \text{threshold} \end{cases}$$

Multi-label :- Noun :- 0.4, Pronoun :- 0.3, Adjective :- 0.05, ...

Threshold :- 0.25 $\rightarrow$  Noun, Pronoun

$\hookrightarrow$ ROC curve $\rightarrow$ Threshold selection.

Softmax Function (Multi-class)

↳ output of a ML algorithm can be a vector of unnormalized scores (logits) for each class. These scores must be converted to probabilities.

Softmax function transforms raw scores (logits) into probabilities that sum up to 1, making it suitable for multiclass classification.

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$$\sum_{i=1}^K \sigma(z_i) = \frac{e^{z_1} + e^{z_2} + \dots + e^{z_K}}{\sum_{j=1}^K e^{z_j}} = 1$$

z_i :- Raw score (logit) for class i

K :- Number of classes

$\sigma(z)_i$:- probability of i^{th} class

eg. {Cat, Dog, Bird}, logits:- $z_{\text{cat}} = 2.0$, $z_{\text{dog}} = 1.0$,
 $z_{\text{bird}} = 0.1$. Find the probabilities of the 3 classes
using the softmax function.

$$\sigma(z)_{\text{cat}} = \frac{e^{2.0}}{e^2 + e^1 + e^{0.1}} = \frac{7.39}{7.39 + 2.72 + 1.11} \approx \boxed{0.659}$$

$$\sigma(z)_{\text{dog}} = \frac{e^1}{e^2 + e^1 + e^{0.1}} \approx \boxed{0.242}$$

$$\sigma(z)_{\text{bird}} = \frac{e^{0.1}}{e^2 + e^1 + e^{0.1}} \approx \boxed{0.099}$$

Sigmoid Function (Binary classification)

For positive class you are getting a single logit (score)

$$\sigma(\text{positive}) = \frac{1}{1 + e^{-z}} \quad \text{logit}$$

eg. $\text{logit} = 1.5$ For spam, what is the probability using sigmoid function?

Sol. $\sigma = \frac{1}{1+e^{-1.5}} \approx 0.817$

Sigmoid Function (Multi-label classification)

For each class i , the sigmoid function o/p the probability between 0 to 1

$$\sigma(z)_i = \frac{1}{1+e^{-z_i}}$$

z_i :- logit of i^{th} class,

$\sigma(z)_i$:- prob of i^{th} class.

eg. we are doing a multi-label classification, with following
cat:- 1.5, Dog:- -0.5, Bird:- 2.0, logits from the
model. what classes does the input label correctly
if we use a threshold probability of 60%.
(Use sigmoid function to calculate probabilities)

Sol. $P(\text{cat}) = \sigma(z)_{\text{cat}} = \frac{1}{1 + e^{-1.5}} \approx 0.817$

$$P(\text{Dog}) = \sigma(z)_{\text{dog}} = \frac{1}{1 + e^{0.5}} \approx 0.37$$

$$P(\text{Bird}) = \sigma(z)_{\text{bird}} = \frac{1}{1 + e^{-2}} \approx 0.88$$

Hence Cat & Bird labels will be given to the
input ($P \geq 0.6$)